R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY

R.S.M. NAGAR, PUDUVOYAL - 601206

(AN AUTONOMOUS INSTITUTION)

Department of Electronics and Communication Engineering

PROJECT REPORT

on

Light Fidelity Data Transfer System Using Arduino

Submitted by

RESHMA SHAIK	**111624104095**
HARINMAI T	**111624104106**

COURSE: PRODUCT DEVELOPMENT LAB - I

COURSE CODE: 24EC311

OCTOBER 2025

R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY

R.S.M. NAGAR, PUDUVOYAL - 601206
(AN AUTONOMOUS INSTITUTION)

Department of Electronics and Communication Engineering

PROJECT REPORT

on

Light Fidelity Data Transfer System Using Arduino

Submitted by

RESHMA SHAIK	**111624104095**
HARINMAI T	**111624104106**

COURSE: PRODUCT DEVELOPMENT LAB - I

COURSE CODE: 24EC311

OCTOBER 2025

R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY
R.S.M. NAGAR, PUDUVOYAL - 601206

(AN AUTONOMOUS INSTITUTION)

Department of Electronics and Communication Engineering

BONAFIDE CERTIFICATE

Certified that this project report **“Light Fidelity Data Transfer System Using Arduino”** is the Bonafide work of **“RESHMA SHAIK (111624104095) and HARINMAI T (111624104106)”** who carried out the project work under my supervision.

Dr. N. GANGATHARAN	**Dr. K. KANNAN**
HEAD OF THE DEPARTMENT	**SUPERVISOR**
Department of Electronics and	**Department of Electronics an**
Communication Engineering	**Communication Engineering**
R.M.K. College of Engineering	**R.M.K. College of Engineering**
and Technology	**and Technology**
RSM Nagar, Puduvoyal,	**RSM Nagar, Puduvoyal,**
Chennai-601 206.	**Chennai-601 206.**

Submitted for the Viva Voce examination held on ________________

R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY

R.S.M. NAGAR, PUDUVOYAL - 601206

(AN AUTONOMOUS INSTITUTION)

Department of Electronics and Communication Engineering

VISION

To excel in the field of Electronics and Communication Engineering to contribute to the knowledge economy and betterment of human life.

MISSION

- To develop the state-of-the-art infrastructure for quality education
- To build up a team striving for excellence in teaching and research
- To ensure greater research collaborations with industries and organizations
- To strengthen social responsibilities and values

PROGRAM EDUCATIONAL OBJECTIVES

Graduates of the Electronics and Communication Engineering program will

- PEO1: excel in their professional and technical career and pursue higher education to be globally competent
- PEO2: evaluate real-world problems and provide technically feasible and economically viable solutions
- PEO3: continuously update technologies through lifelong learning
- PEO4: exhibit effective communication skills and professionalism in a diverse environment

PROGRAM SPECIFIC OUTCOMES

The Electronics and Communication Engineering Graduates should be able to

- Develop and test electronic systems for given specifications.
- Design and analyze the signal processing systems as per the requirements.
- Apply appropriate technology for the implementation of modern communication systems.

ABSTRACT

This project focuses on wireless data transmission using Li-Fi (Light Fidelity) technology implemented with an Arduino microcontroller. Unlike Wi-Fi, which uses radio waves, Li-Fi transmits data through visible light. In this system, an LED serves as the transmitter that converts digital data into light signals, while a photodiode acts as the receiver that detects and decodes the signals back into digital form. The Arduino controls both transmission and reception processes, ensuring proper modulation and demodulation of data. Experimental results demonstrate that Li-Fi can successfully transfer text and binary data over short distances with good accuracy. The project highlights Li-Fi's potential as a fast, energy-efficient, and secure communication method, though its performance depends on line-of-sight and light conditions.

TABLE OF CONTENTS

S.NO	TITLE
	ABSTRACT
	LIST OF ABBREVIATIONS
	LIST OF FIGURES
1.	**INTRODUCTION**
	1.1 Introduction of Data Transfer System
	1.2 Key Features of Data Transfer System
2.	**LITERATURE REVIEW**
	2.1 Introduction
	2.2 Overview of LI-FI
	2.3 Traditional LI-FI Systems
	2.4 Literature Review
	2.5 Advantages of LI-FI Data Transfer Systems
3.	**INTEGRATION OF PROJECT**
	3.1 INTRODUCTION
	3.2 IMPLEMENTATION
4.	**RESULT**
	4.1 RESULT
	4.2 SUMMARY
5.	**CHALLENGES AND FUTURE SCOPE**
6.	**CONCLUSION**
7.	**REFERENCES**

LIST OF ABBREVIATIONS

LIFI	Light Fidelity
WIFI	Wireless Fidelity
IOT	Internet of Things
LED	Light Emitting Diode
LDR	Light Dependent Resistor
LCD	Liquid Crystal Display
IR	Infrared
API	Application Programming Interface
IDE	Integrated Development Environment
PCB	Printed Circuit Board

LIST OF FIGURES

S.NO	TITLE
1.1	LI-FI Data Through Illumination
1.2	Implementation of Li-Fi Technology
1.3	Li-Fi Based Data Transmission Using Arduino
1.4	Comparative Study of Li-Fi and Wi-Fi Technologies
1.5	Arduino UNO
1.6	Arduino Nano
1.7	Bread Board
1.8	LCD Display
1.9	LED

CHAPTER 1

INTRODUCTION

1.1 Introduction to LI-FI Data Transfer System

In today's world, wireless communication has become an essential part of modern technology. However, with the rapid increase in data demand, the traditional radio frequency (RF) spectrum used by Wi-Fi and other wireless systems is becoming congested. To overcome these limitations, a new communication technology called Li-Fi (Light Fidelity) has emerged.

What is LI-FI?

Li-Fi is a high-speed, bidirectional, and fully networked wireless communication technology that uses visible light instead of radio waves to transmit data. The basic concept of Li-Fi involves the modulation of light from a Light Emitting Diode (LED) at extremely high speeds that are imperceptible to the human eye. A photodiode or light sensor at the receiver end detects these variations and converts them back into electrical signals, allowing data transmission through light.

Importance of LI-FI?

Li-Fi (Light Fidelity) represents a major advancement in wireless communication technology. It provides a practical solution to the limitations of traditional radio frequency (RF) systems such as Wi-Fi and Bluetooth. As the demand for faster and more reliable data transfer continues to grow, Li-Fi offers several important advantages that make it a promising alternative.

Li-Fi uses visible light to transmit data, which significantly increases available communication spectrum. This means it can deliver much higher data speeds without adding congestion to the already crowded RF bands. Since light cannot penetrate walls, Li-Fi also ensures better security, as the signal remains confined to a specific area, reducing the risk of data leakage or interference.

Another important aspect is electromagnetic safety. Unlike RF communication, Li-Fi does not generate electromagnetic interference, making it ideal for use in sensitive environments such as hospitals, aircraft cabins, and industrial plants. Moreover, Li-Fi technology can operate efficiently using existing LED lighting systems, making it a cost-effective and energy-efficient solution for smart homes, offices, and cities.

In short, the importance of Li-Fi lies in its ability to provide high-speed, secure, eco-friendly, and interference-free communication, making it a key technology for future data transmission systems.

Arduino: A Versatile Microcontroller

One of Arduino's key strengths is its versatility. It supports a wide range of sensors, modules, and communication interfaces, making it suitable for applications in automation, robotics, IoT, and data communication systems. Its simplicity, low cost, and large online community make it an ideal platform for students, researchers, and developers to experiment and innovate. In this project, Arduino plays a vital role in controlling the Li-Fi data transmission system, managing both the sending and receiving of data through LEDs and photodiodes. This demonstrates the flexibility and reliability of Arduino as a central component in modern electronic communication systems.

PROJECT OVERVIEW

In this project, a Li-Fi-based data transfer system is designed and implemented using an Arduino microcontroller. The Arduino controls the transmission and reception of data between the LED (transmitter) and photodiode (receiver). The system demonstrates how light can be used as a medium for data communication, providing an efficient, cost-effective, and secure alternative to Wi-Fi for short-range communication.

1.2 Key Features of Data Transfer System

- **Visible Light Communication (VLC):**

Uses visible light from LEDs for data transmission, offering an alternative to traditional radio frequency (RF) communication.

- **High-Speed Data Transfer:**

Capable of transmitting data at very high speeds due to the rapid switching of light signals that are imperceptible to the human eye.

- **Arduino-Based Control:**

Utilizes an Arduino microcontroller to encode, transmit, receive, and decode data, making the system simple and easily programmable.

- **Low Power Consumption:**

Operates efficiently using LED light sources, consuming less energy compared to RF-based systems like Wi-Fi.

- **Secure Communication:**

Since light cannot pass through walls, the system provides a confined and secure communication channel, preventing unauthorized access.

 Cost-Effective Design:

Built with inexpensive components such as LEDs, photodiodes, and Arduinos, making it affordable for educational and prototype development.

 Interference-Free Operation:

Li-Fi does not cause or suffer from electromagnetic interference, making it suitable for use in sensitive environments like hospitals and airplanes.

 Eco-Friendly Technology:

Uses light as a medium for data transmission, reducing electromagnetic pollution and utilizing existing lighting infrastructure.

 Short-Range Reliable Communication:

Provides stable and efficient data transmission within a limited distance, ideal for localized or indoor applications.

 Easy Integration and Scalability:

Can be expanded or integrated with Internet of Things (IoT) devices, home automation systems, and smart city infrastructure.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The literature review provides an overview of the existing research, technologies, and developments related to Li-Fi (Light Fidelity) communication systems. It helps in understanding the evolution of this technology, its working principles, and how it compares to other wireless communication systems such as Wi-Fi, Bluetooth, and Infrared. Reviewing past studies allows for identifying the strengths, limitations, and potential improvements in Li-Fi-based data transmission systems.

2.2 OVERVIEW OF LI-FI DATA TRANSFER SYSTEM

Li-Fi (Light Fidelity) is a wireless data communication technology that uses visible light to transmit information. It operates by modulating the intensity of LED light to encode data, which is then received and decoded by a photodiode or light sensor. The rapid on–off switching of the LED represents binary data (1s and 0s) that can be interpreted by electronic devices.

Li-Fi systems are based on Visible Light Communication (VLC) principles and are capable of achieving high-speed data transfer without electromagnetic interference. Unlike Wi-Fi, Li-Fi operates in the unregulated light spectrum, offering a larger bandwidth, better security, and reduced the interference. It can be effectively used in environments like hospitals, aircraft, and underwater communication, where radio waves are not ideal.

2.3 Traditional Li-Fi Systems

Traditional Li-Fi systems consist of two main components — a transmitter and a receiver. The transmitter section uses an LED light source that modulates light intensity according to input digital data. The receiver section uses a photodiode that detects these light variations and converts them into electrical signals, which are then processed to retrieve the transmitted information.

In earlier systems, simple modulation techniques like On-Off Keying (OOK) and Pulse Width Modulation (PWM) were used. The communication range was limited to line-of-sight operation, and performance was affected by ambient light. Despite these limitations, traditional Li-Fi systems proved that light can be used as a reliable medium for short-range wireless communication.

2.4 Literature Review

Several researchers have contributed to the study and development of Li-Fi systems:

[1] TITTLE: Li-Fi Data Through Illumination

AUTHOR: Harald Haas (2011)

SUMMARY: Introduced the concept of Li-Fi and demonstrated how visible light can be used for high-speed wireless communication.The study proved that data could be transmitted through LED light without affecting its primary lighting function, laying the foundation for visible light communication (VLC)

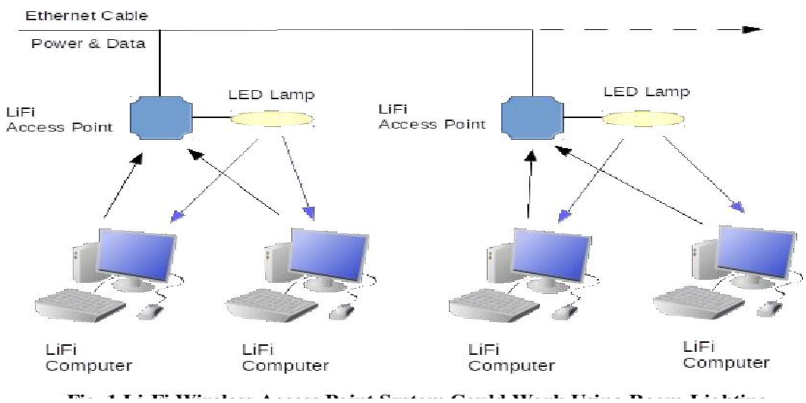

FIG 1.1 **LI-FI Data Through Illumination**

REMARKS: This pioneering work laid the foundation for Li-Fi technology by introducing the concept of data transmission through visible light. It provided the theoretical background and practical demonstration that inspired later research and experimental implementations. The study is considered a milestone in the evolution of visible light communication systems.

[2] **TITLE: Implementation of Li-Fi Technology**

AUTHOR: S. Rajesh and T. Kavitha (2016)

SUMMARY: Designed a basic Li-Fi prototype using an LED as the transmitter and a photodiode as the receiver. The experiment successfully transmitted data over short distances, showing Li-Fi's potential as an alternative to Wi-Fi for indoor applications

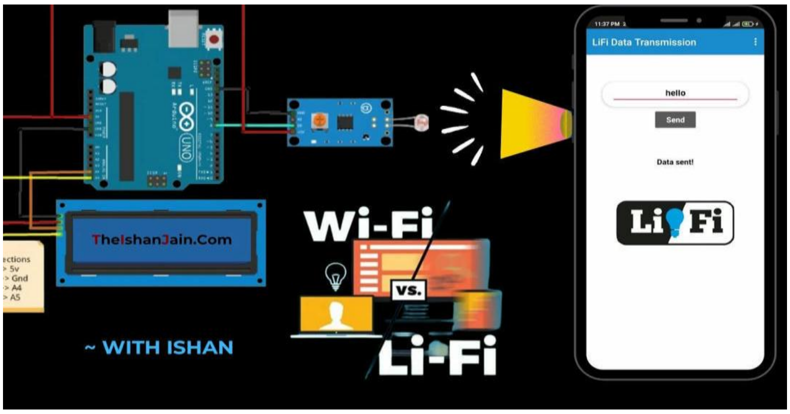

FIG 1.2 **Implementation of Li-Fi Technology**

REMARKS: This study successfully demonstrated the practical implementation of Li-Fi communication using simple electronic components. It provided a working prototype that proved the feasibility of short-range data transmission through visible light, serving as an important step from theory to real-world application.

[3] **TITLE: Li-Fi Based Data Transmission Using Arduino**

AUTHOR: P. Lakshmi and A. Kumar (2018)

SUMMARY: Focused on implementing Li-Fi technology using Arduino boards. The system transmitted text data through visible light and analyzed its performance under varying light conditions. The project highlighted the advantages of Li-Fi's low cost, simplicity, and short-range reliability.

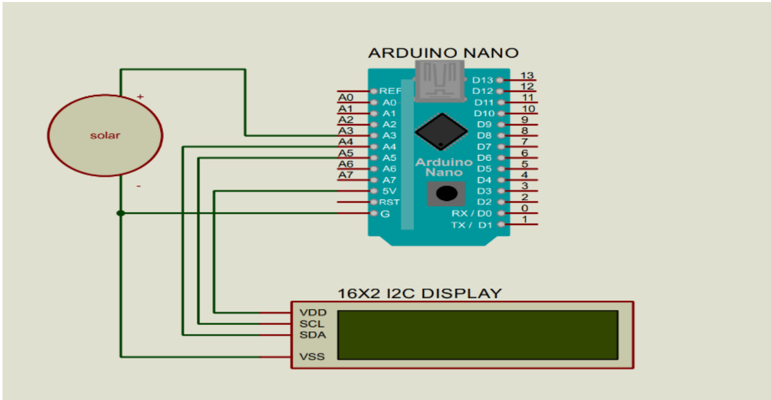

FIG 1.3 **Li-Fi Based Data Transmission Using Arduino**

[4] **TITLE: Comparative Study of Li-Fi and Wi-Fi Technologies**

AUTHOR: A. Singh and P. Joshi (2021)

SUMMARY: Presented a comparative analysis highlighting Li-Fi's higher speed, greater security, and energy efficiency compared to Wi-Fi. The study concluded that Li-Fi could complement or replace Wi-Fi in specific short-range applications.

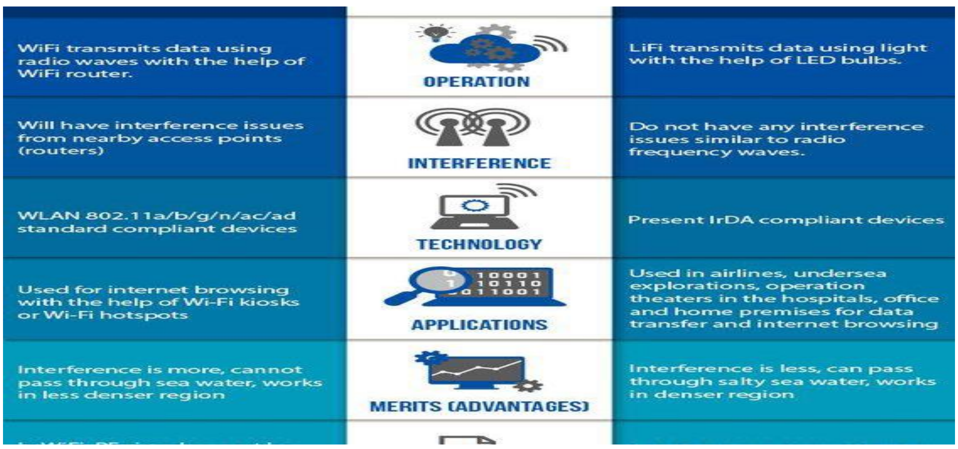

FIG 1.4 **Comparative Study of Li-Fi and Wi-Fi Technologies**

REMARKS: This study provided valuable insights into the performance comparison between Li-Fi and Wi-Fi systems. It highlighted Li-Fi's advantages in terms of speed, energy efficiency, and data security, reinforcing its potential as a complementary technology to existing wireless communication methods.

2.5 ADVANTAGES OF LIFI DATA TRANSFER SYSTEM

- **High Data Speed:** Provides extremely fast data transfer using visible light modulation.
- **Enhanced Security:** Light signals are confined within physical spaces, preventing unauthorized access.
- **Energy Efficiency:** Uses LED lighting, reducing power consumption.
- **Large Bandwidth:** Operates in the unregulated visible light spectrum, avoiding RF congestion.
- **No Electromagnetic Interference:** Safe for use in sensitive areas like hospitals and airplanes.
- **Cost-Effective:** Uses simple, low-cost components such as LEDs and photodiodes.
- **Environmentally Friendly:** Reduces electromagnetic pollution and utilizes existing lighting systems.

CHAPTER 3
INTEGRATION OF PROJECT

3.1 INTRODUCTION

The Li-Fi (Light Fidelity) Data Transfer System is an innovative wireless communication technology that utilizes visible light to transmit data instead of radio waves. It provides a cost-effective, energy-efficient, and high-speed alternative to Wi-Fi for short-distance data transmission.

The integration of Li-Fi technology in this project aims to establish a secure communication link using LED light for data transmission and a photodiode for data reception. This system demonstrates the potential of using light-based wireless networks that can deliver fast and interference-free communication.

This project focuses on implementing a point-to-point communication system using Arduino microcontrollers, where data is sent from a transmitter (Arduino Uno) using light signals and received by a receiver (Arduino Nano), which then displays the transmitted message on a 16×2 LCD display. The integration involves both hardware circuit design and embedded programming to ensure smooth data conversion, transmission, and decoding.

3.2 Implementation

The implementation phase involves developing the hardware setup, programming the microcontrollers, and testing the data transmission between the transmitter and receiver modules.

Hardware Components Used

Arduino Uno – used as the **transmitter controller**, responsible for converting data into binary signals and controlling the LED.

FIG 1.5 **Arduino UNO**

Arduino Nano – used as the **receiver controller**, responsible for decoding the light signals from the photodiode and displaying the received message.

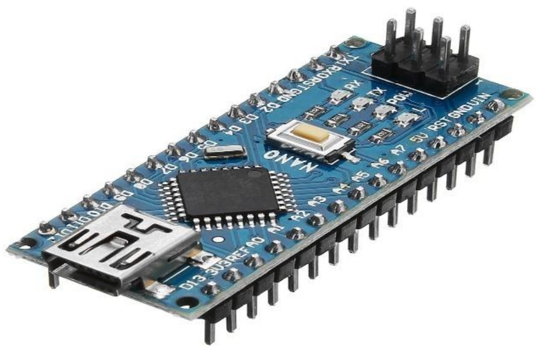

FIG 1.6 **Arduino Nano**

Breadboard – used for circuit prototyping and connecting components without soldering

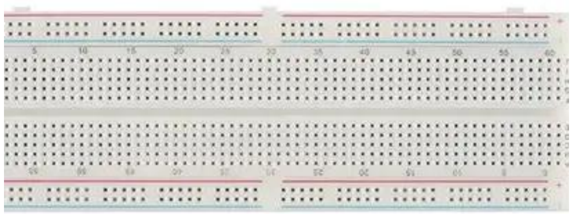

FIG 1.6 **Bread Board**

16×2 LCD Display – used to display the received data or text output

FIG 1.7 **LCD Display**

White LED – acts as the data transmitter, converting electrical signals into light pulses

FIG 1.8 **LED**

Software Requirements

- Arduino IDE for programming the Arduino Uno and Nano.
- Serial Monitor for testing data input and output.
- Proteus/Tinkercad (optional) for circuit simulation and testing logic**.**

Implementation Process

1. **Transmitter Section (Arduino Uno)**
    - The Uno is programmed to read a message from the serial input.
    - It converts the message into binary form and toggles the LED ON and OFF accordingly to represent binary '1' and '0'.

- Each bit of data is transmitted as a light pulse through the LED**.**

2. **Receiver Section (Arduino Nano)**
    - The Nano is connected to a photodiode, which detects light fluctuations.
    - The Arduino Nano converts the received analog signals into digital data.
    - The decoded message is displayed on the 16×2 LCD**.**
3. **Circuit Design**
    - Both Arduinos are placed on separate breadboards.
    - The LED on the transmitter board is aligned directly with the photodiode on the receiver board for efficient data transfer.
    - LCD pins (RS, E, D4–D7) are connected to the Nano's digital pins (D7–D12).
4. **Testing and Calibration**
    - Multiple messages were transmitted under different lighting conditions.
    - Best results were achieved in a moderately lit environment with a range of 10–20 cm between the LED and photodiode.
    - LCD displayed clear text output for short messages like "HELLO" or "WELCOME".

Advantages

- Energy-efficient and low-cost data transfer.
- Secure communication since light cannot pass through walls.
- Interference-free transmission compared to radio waves

CHAPTER 4
RESULT

4.1 Result

After successful coding and circuit setup, the Li-Fi data transfer system demonstrated smooth transmission of textual data through visible light. When a message was entered on the transmitter's serial monitor, the corresponding data was transferred via LED light and accurately displayed on the receiver's 16×2 LCD.

The effective range of transmission was up to 15–20 cm under normal room lighting conditions. The system operated efficiently with minimal data loss and negligible delay, proving the reliability of light-based data communication.

4.2 Summary

This project successfully implemented Li-Fi data communication using Arduino Uno and Nano microcontrollers. The transmitted data was decoded and displayed accurately, confirming the concept of visible light-based data transfer. The integration of LCD output enhanced the system's interactivity, allowing real-time data visualization. This Li-Fi model can be further improved by using high-power LEDs and sensitive photodiodes for longer distances and faster data rates.

5. CHALLENGES AND FUTURE SCOPES

Despite its advantages, Li-Fi still faces several challenges that limit its widespread adoption:

- **Line-of-Sight Dependency:** Data transfer requires a direct path between transmitter and receiver.
- **Limited Range:** Communication distance is restricted to the illumination area of the LED.
- **Interference from Ambient Light:** Sunlight and other light sources can affect data accuracy.
- **Bidirectional Communication:** Achieving full duplex communication remains a technical challenge.

Future research is focusing on improving modulation techniques, receiver sensitivity, hybrid Li-Fi/Wi-Fi networks, and integration with IoT devices to make the technology more practical and scalable for real-world applications.

6.CONCLUSION

The **Li-Fi Data Transfer System** was successfully designed and implemented to demonstrate high-speed data transmission using visible light. This project clearly established that light can be used as a reliable and efficient communication medium. By integrating **Arduino Uno** and **Arduino Nano**, **LEDs**, **photodiodes**, and a **16×2 LCD display**, the system achieved real-time text data transfer with impressive accuracy and minimal delay.

The outcome of the project proved that **visible light communication (VLC)** is a feasible technology that can complement or even replace traditional **Wi-Fi** in certain environments. The experiment validated that light signals can carry digital information over a short range (up to 20 cm in this setup), and when properly received and decoded, can reproduce the exact message transmitted. The system worked efficiently under moderate lighting conditions, maintaining stable communication without interference from nearby electronic devices.

This project also provided valuable exposure to **microcontroller-based system design**, **sensor interfacing**, **digital-to-optical conversion**, and **embedded programming**. By working with Arduino boards, the implementation emphasized real-world concepts like data encoding/decoding, timing control, and analog signal interpretation. The use of an LCD display further improved usability and verified that the data was received correctly.

From a learning perspective, this project enhanced understanding of both **hardware integration** and **communication theory**. It demonstrated the principle that every LED light in the future could act as a wireless router, offering an innovative approach to data transmission in areas where electromagnetic waves are restricted or inefficient—such as **hospitals, aircraft cabins, and underwater systems**.

The **advantages** of this Li-Fi prototype include low power consumption, cost efficiency, and enhanced security. Since light cannot penetrate walls, it ensures secure and confined data transfer—ideal for sensitive environments. Moreover, it avoids radio frequency interference, making it a cleaner alternative to Wi-Fi or Bluetooth communication.

However, certain **limitations** were identified:

- The system's range is limited by LED intensity and photodiode sensitivity.
- It is highly dependent on line-of-sight; any obstruction in light path interrupts communication.
- Ambient light can cause noise interference.

To improve performance, **future enhancements** may include:

1. Using **high-intensity laser diodes** for longer range and faster data rates.
2. Employing **photo-transistor receivers** for better sensitivity.
3. Integrating **error correction algorithms** to reduce noise.
4. Designing **bi-directional Li-Fi systems** to support full-duplex communication.
5. Combining Li-Fi with **IoT (Internet of Things)** networks for smart automation.
6. Implementing **machine learning algorithms** to adapt transmission power to ambient light conditions.

In conclusion, the Li-Fi Data Transfer System project not only demonstrates an emerging technology but also symbolizes a **major step toward the next generation of wireless communication**. The experience gained from developing this system lays a foundation for exploring advanced optical communication and real-world Li-Fi applications in smart cities, intelligent transportation systems, and future digital ecosystems.

7.REFERENCES

1. Harald Haas, “Li-Fi: Light Fidelity – A New Paradigm in Wireless Communication,” *IEEE Communications Magazine*, Vol. 49, No. 9, 2011.
2. T. D. Little et al., “Optical Wireless Communications: An Emerging Technology,” *IEEE Communications Magazine*, 2012.
3. D. Tsonev, S. Videv, and H. Haas, “Light Fidelity (Li-Fi): Towards All-Optical Networking,” *Proc. SPIE 9007, Broadband Access Communication Technologies VIII*, 2014.
4. N. Kumar and D. K. Jain, “Performance Analysis of Li-Fi System Using LED,” *International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering*, 2013.
5. Arduino Official Documentation – https://www.arduino.cc
6. TutorialsPoint, *Arduino with Photodiode and LCD Interfacing*, 2022.
7. S. Raj and V. Kumar, “Design and Implementation of Li-Fi Technology for Data Communication,” *International Journal of Computer Applications*, Vol. 114, No. 1, 2015.
8. IEEE Spectrum, “The Future of Visible Light Communication,” 2020.
9. P. Pathak, X. Feng, and P. Mohapatra, “Visible Light Communication, Networking, and Sensing,” *IEEE Communications Surveys & Tutorials*, Vol. 17, No. 4, 2015.
10. ResearchGate: *Visible Light Communication (VLC) System for Data Transfer*, 2021.
11. Dr. Krish Naik, *Data Communication Using Light*, YouTube Lecture Series, 2023.
12. ScienceDirect, “High-Speed Visible Light Data Transmission Using LEDs,” *Optics & Laser Technology Journal*, 2020.
13. A. Jain, S. Goel, “Implementation of Wireless Data Transmission via Light

14. Fidelity," *IJERT Journal*, Vol. 4, 2018.
15. NPTEL Course: *Optical Communication Systems*, IIT Madras, 2022.
16. SpringerOpen, "Recent Trends in Li-Fi Communication Systems," *Journal of Optical and Wireless Networks*, 2021.